Lecture 13:

VLSI Clock Phase Detector Circuits

Contents

This Topic: VLSI Phase Detector Circuits

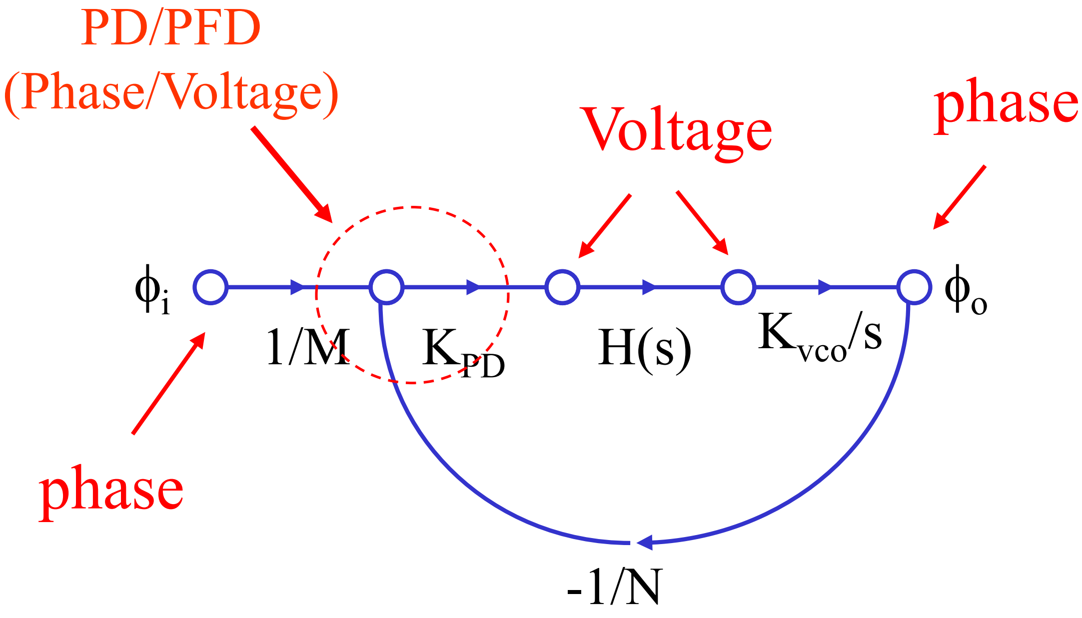
- Introduction
- VLSI Clock Phase Detector Circuits
- Frequency/Phase Detector Circuits
- VLSI CPD Circuit Implementations

Introduction

- VLSI phase detector (PD) including both clock PD (CPD) and data PD (DPD) are used to compare two TD signals
- Both inputs of the CPD are typically periodic (clock) signals of same frequency
- The output of CPD circuits are discrete-time signal of voltage or current, in either digital or analog forms
- The average of the CPD outputs over the reference clock period is a voltage (or current) signal that is proportional to the delay (or phase) difference of the two TD signals
- VLSI CPD circuits are commonly used in PLL, DLL and other time-domain circuit applications

VLSI CPD Circuit Applications

- Example: clock phase detectors (CPDs) are essential circuits in the VLSI PLL circuits
  - Phase comparison
  - Phase to Voltage conversion

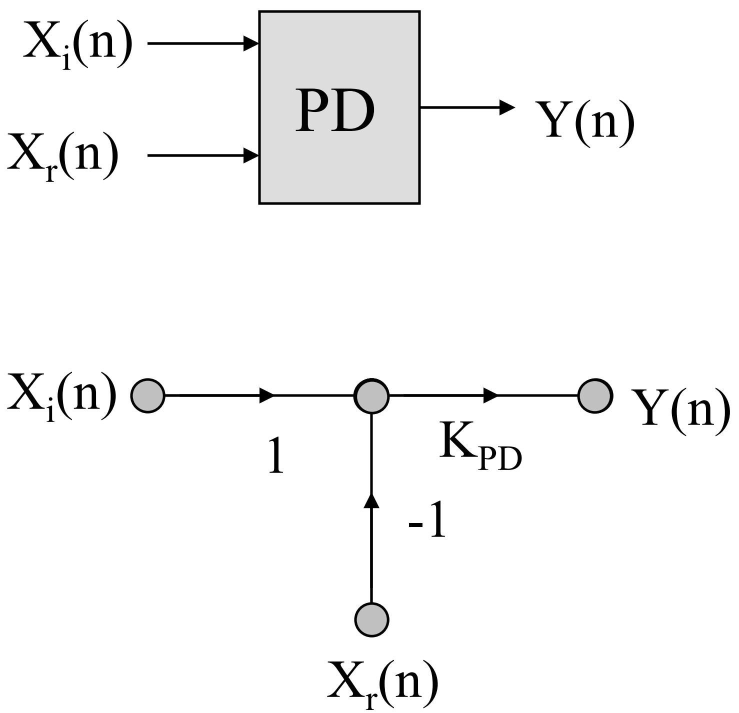


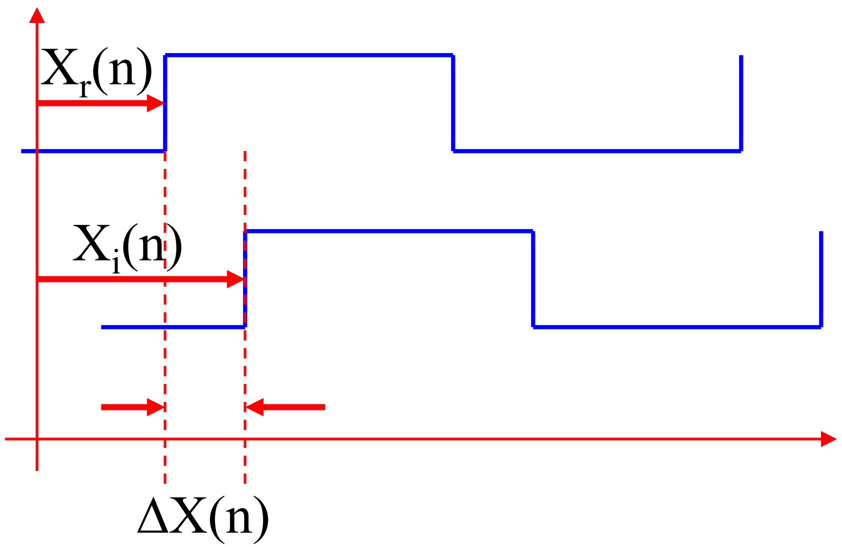
Basic VLSI CPD Operation

Voltage (current) signal

Time domain signals

$$\overline{Y(n)} = K_{PD}(X_i(n) - X_r(n))$$





Note: DP circuit realize Time-Domain to Voltage (current) conversion

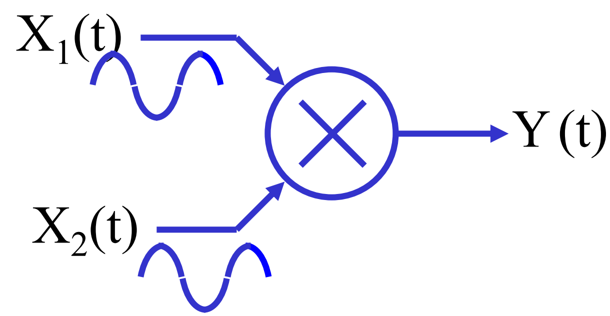
VLSI CPD Circuit Architectures

- Analog Multiplier based CPD
- AND based CPD
- Gilbert Cell based CPD
- XOR based CPD
- S-R Latch based CPD
- Phase/Frequency CPD
- TDC

Analog Multiplier CPD

- Analog multiplication of two input sinusoidal clocks

$$\begin{cases} x_1(t) = A_1\cos(\omega_1 t) \\ x_2(t) = A_2\cos(\omega_2 t + \Delta\phi) \\ y(t) = x_1(t)\cdot x_2(t) \end{cases}$$

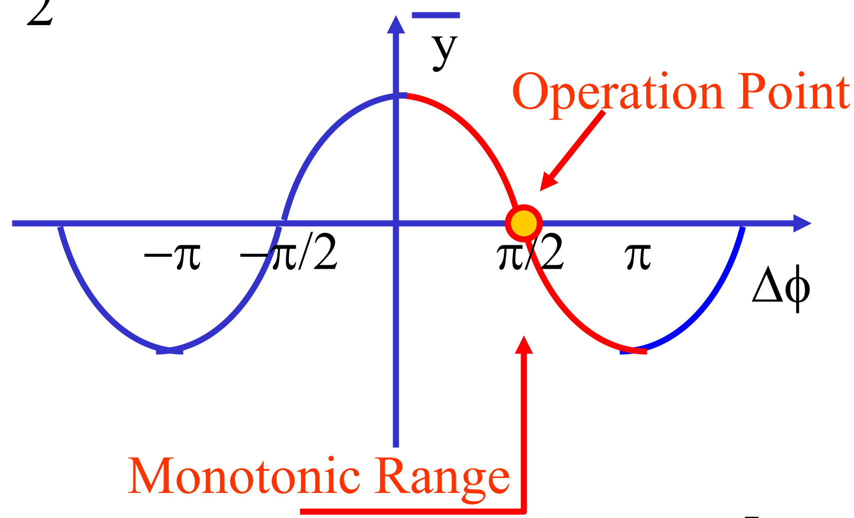


$$y(t) = \frac{A_1A_2}{2}\cos((\omega_1+\omega_2)t+\Delta\phi) + \frac{A_1A_2}{2}\cos((\omega_1-\omega_2)t+\Delta\phi)$$

IF $\omega_1 = \omega_2$ and after averaging:

$$\boxed{\overline{y(t)} = \frac{A_1A_2}{2}\cos(\Delta\phi)}$$

Detection Gain

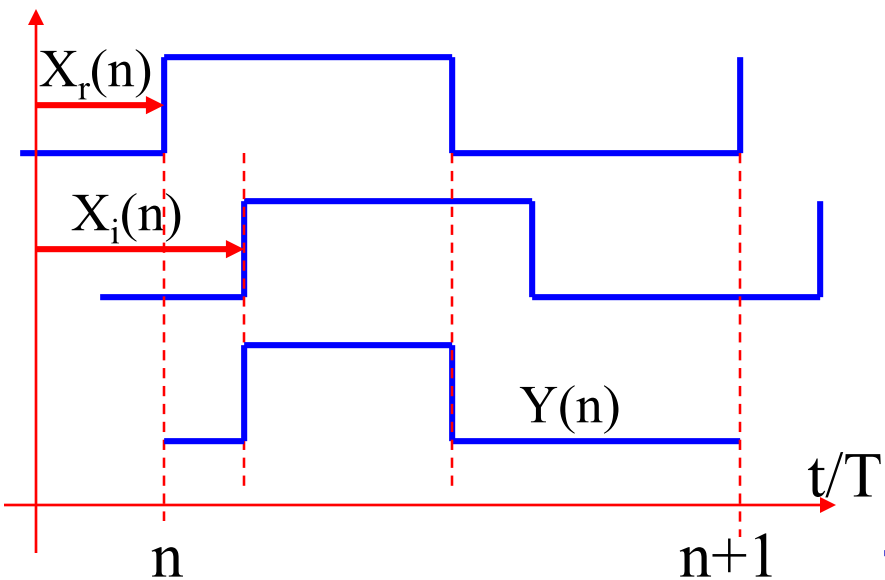


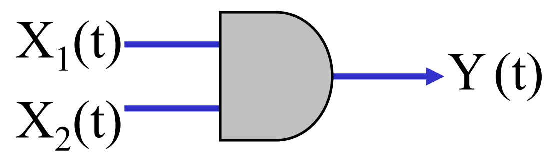
Characteristics of Analog Multiplier CPD

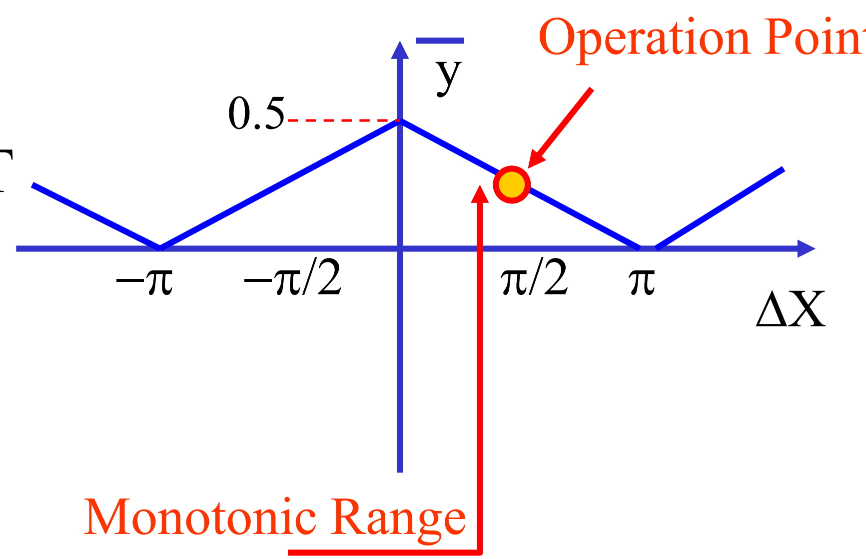
- A traditional phase detection technique
  - Analog based circuit implementation
- Non-linear gain
  - ($K_{pd} \sim \cos(\Delta\omega)$)
- 90^{o} inherent phase shift
  - Output proportional to $\cos(\Delta\omega)$
  - Two inputs are not aligned at locking condition
- Not suitable for frequency detection
  - (Must $\omega_1 = \omega_2$)
  - No harmonic detection capability
- Poor AM noise rejection
  - ($K_{pd} \sim A_1 A_2$)

AND Based CPD

- Digital "multiplication" of two single-end digital clock inputs







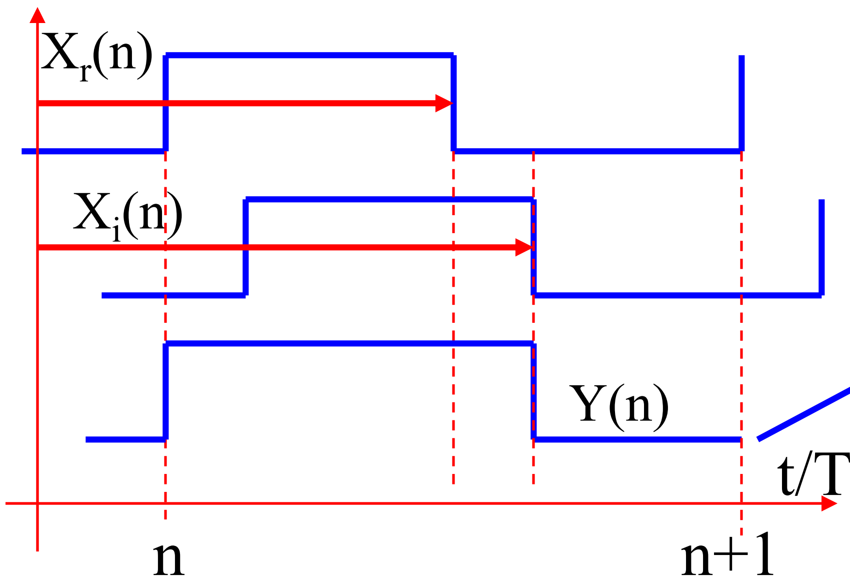
$$\overline{y(n)} = \left|\frac{1}{2} - \frac{\Delta X(n)}{T}\right|$$

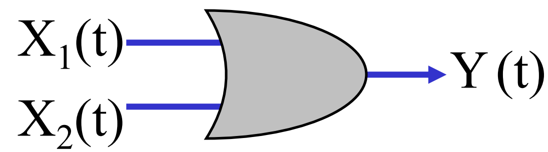
Characteristics of AND CPD

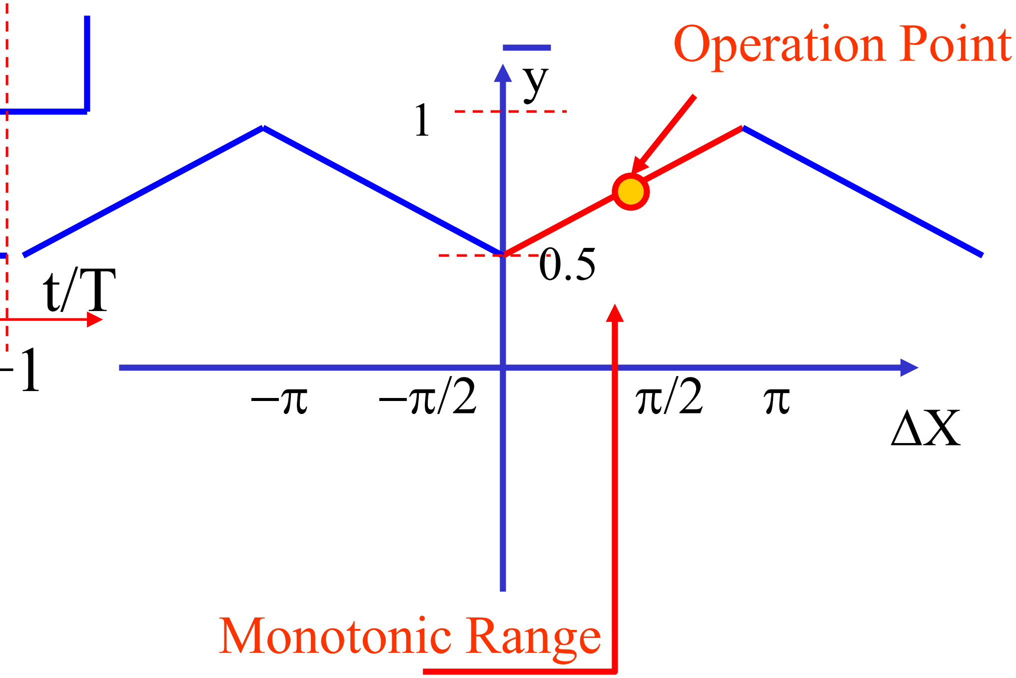
- Fully digital based implementation
  - Better AM noise rejection
- Very linear gain
- 90^o inherent phase shift
  - Two inputs are not aligned during locking
- Requires a DC voltage offset to work as monotonic range
  - Sensitive to duty cycle distortion
- Not suitable for frequency detection
  - (Must $\omega_1 = \omega_2$)
  - No harmonic detection capability

OR Based CPD circuits

- Alternative CPD solution: Digital “addition” of two inputs







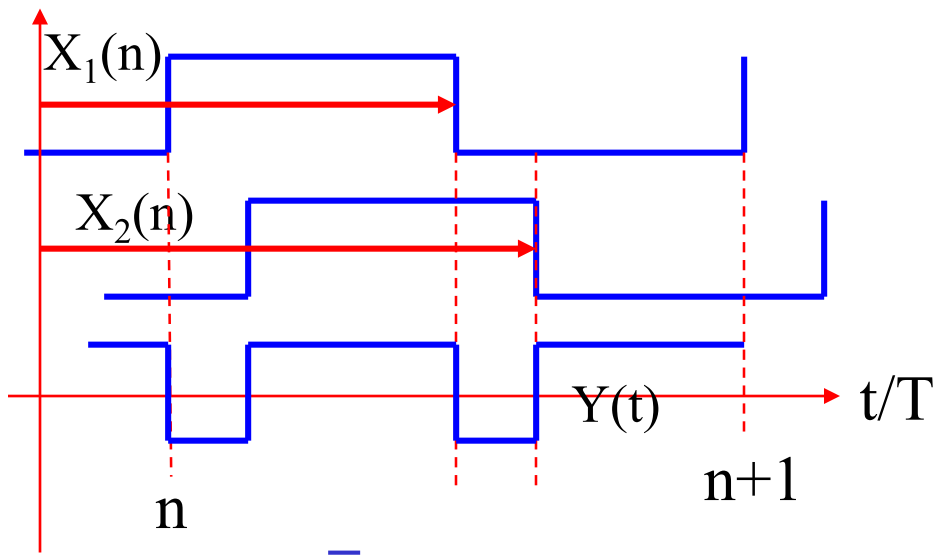
$$\overline{y(n)} = \left|\frac{1}{2} + \frac{\Delta X(n)}{2}\right|$$

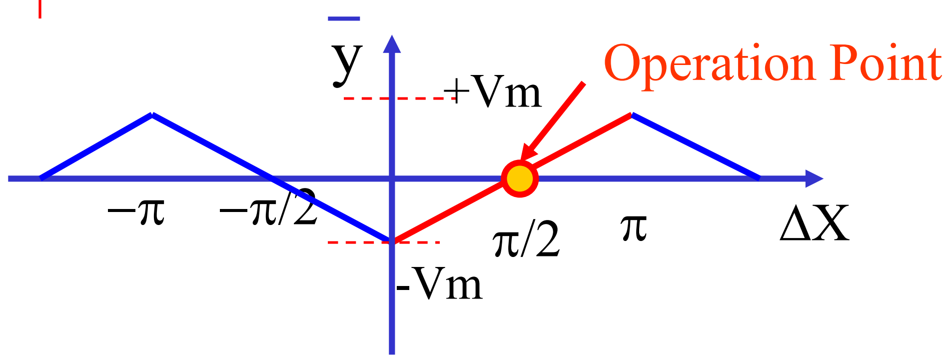
Characteristics of OR CPD

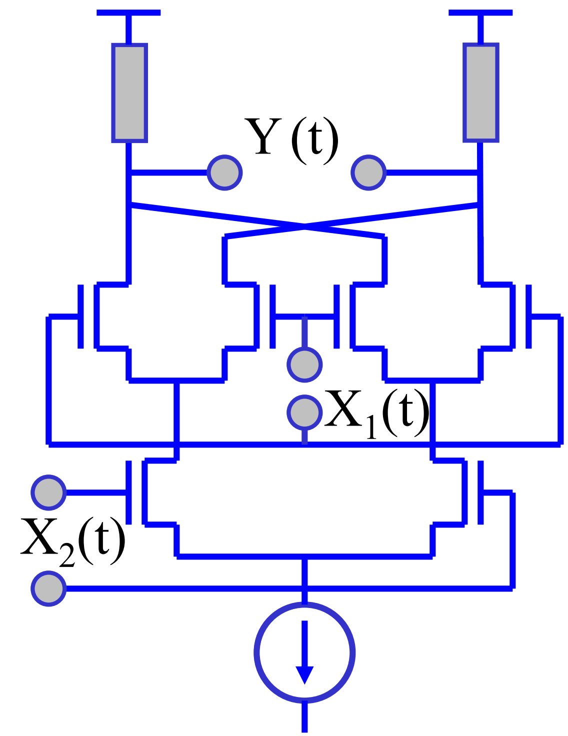
- Fully digital based implementation
  - Better AM noise rejection
- Very linear gain
- 90^o inherent phase shift
  - Two inputs are not aligned during locking
- Requires a DC voltage offset to work as monotonic range
  - Sensitive to duty cycle distortion
- Not suitable for frequency detection
  - (Must $\omega_1 = \omega_2$)
  - No harmonic detection capability

Gilbert Cell CPD

- Digital multiplication of two differential input clocks



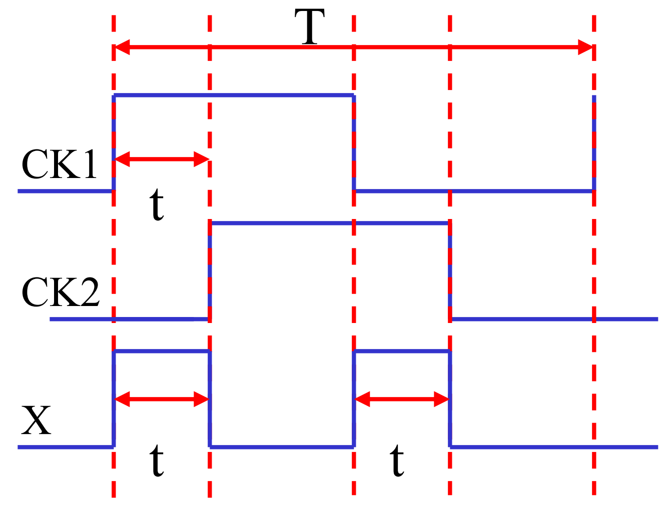


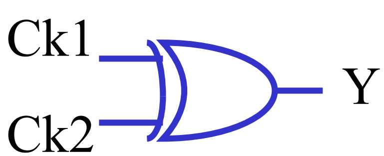


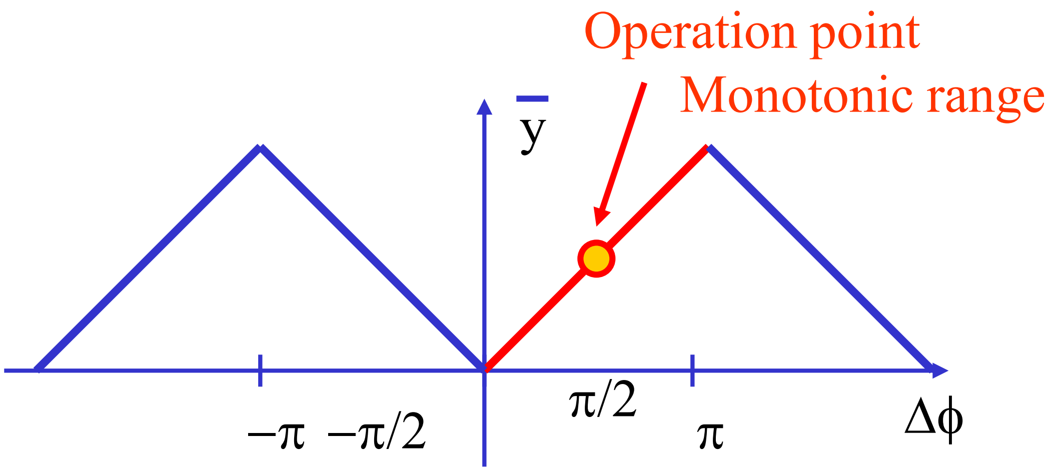
$$\overline{y(n)} = \frac{\Delta X(n)}{\pi} - \frac{1}{2}$$

XOR-Based CPD

- Combinational digital operation
- Better AM rejection
- Constant gain
- Duty-cycle dependent
- Not for frequency detection
- High rejection to AM noise



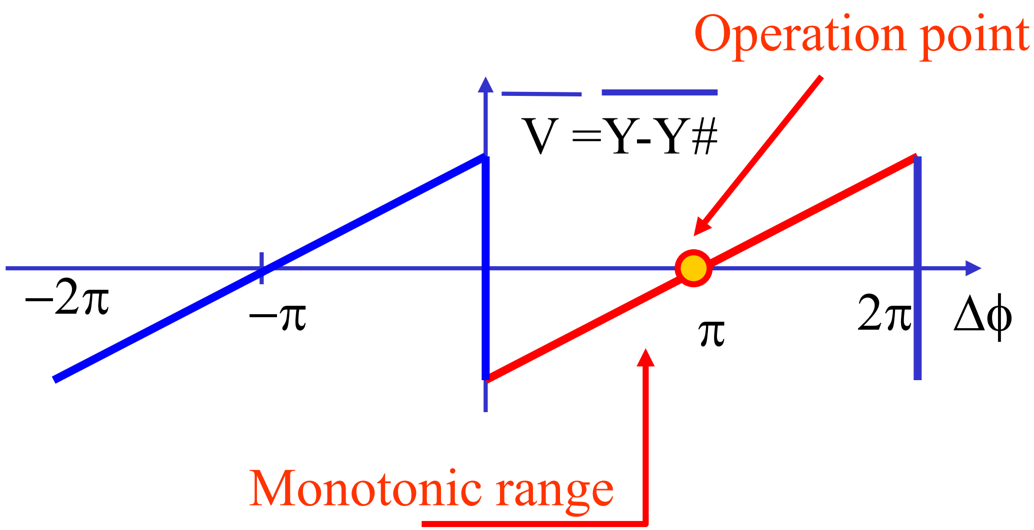


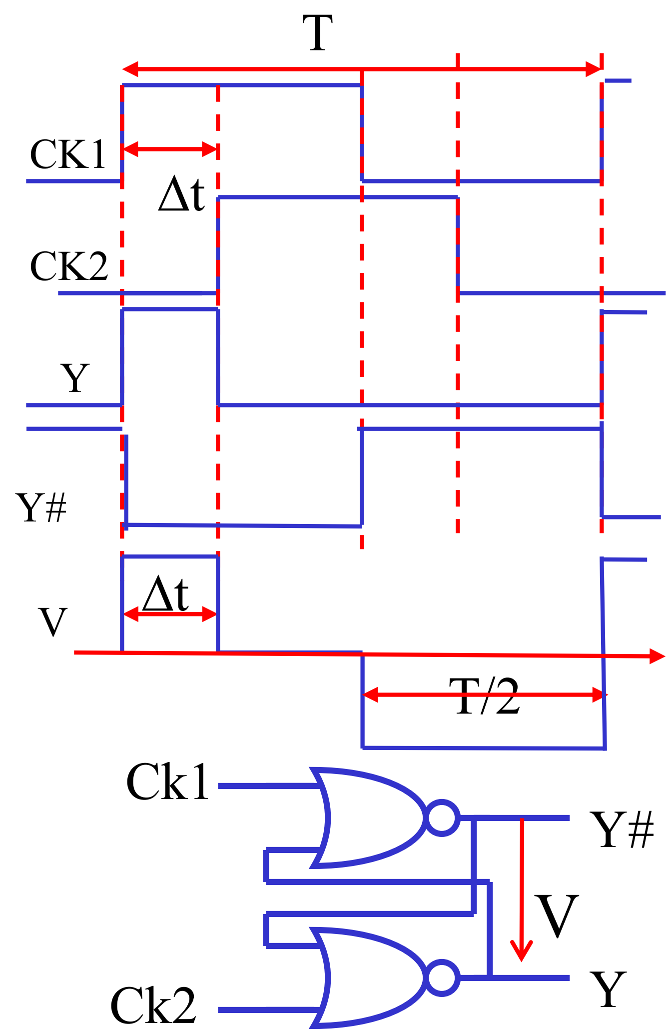


$$\overline{y(n)} = \frac{\Delta X(n)}{2\pi}$$

R-S Latch-Based CPD

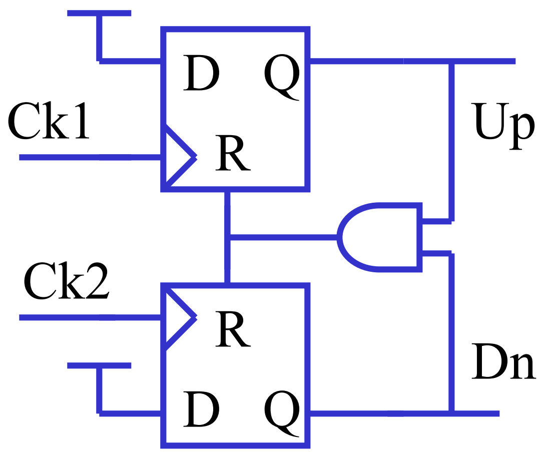
- Edge triggered sequential circuit
- Output frequency same as input
- Duty-cycle independent
- Not for frequency detection
  - Harmonic locking problem

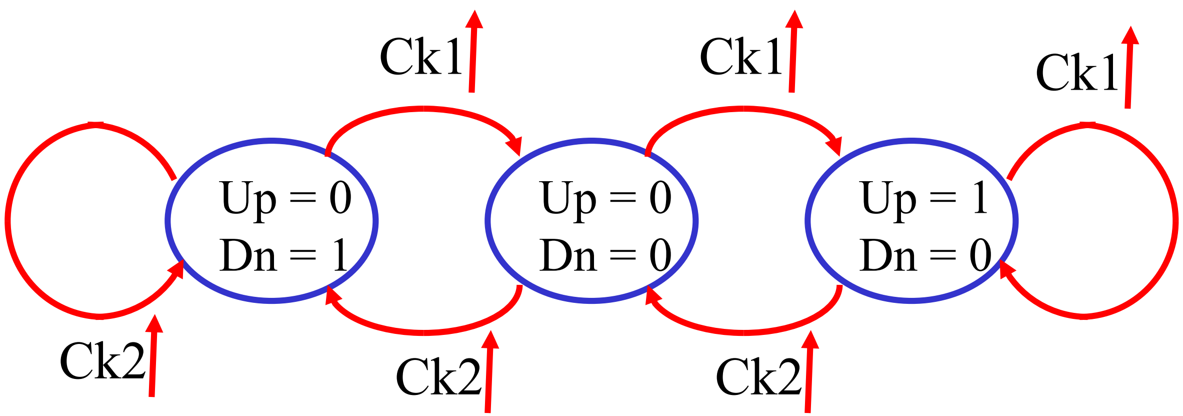




Phase Frequency CPD

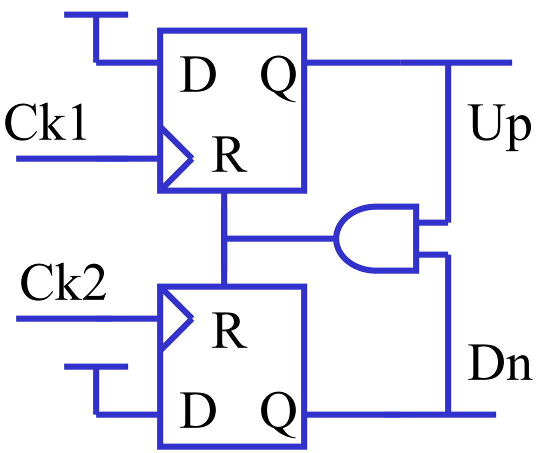
- Three-state operation
  - Up = 0, Dn = 0
  - Up = 1, Dn = 0
  - Up = 0, Dn = 1
- Edge-triggered

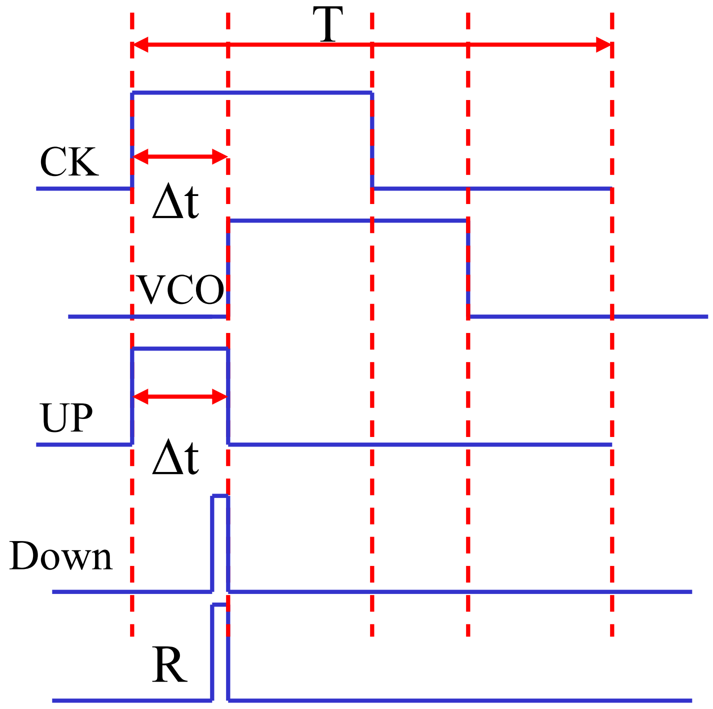


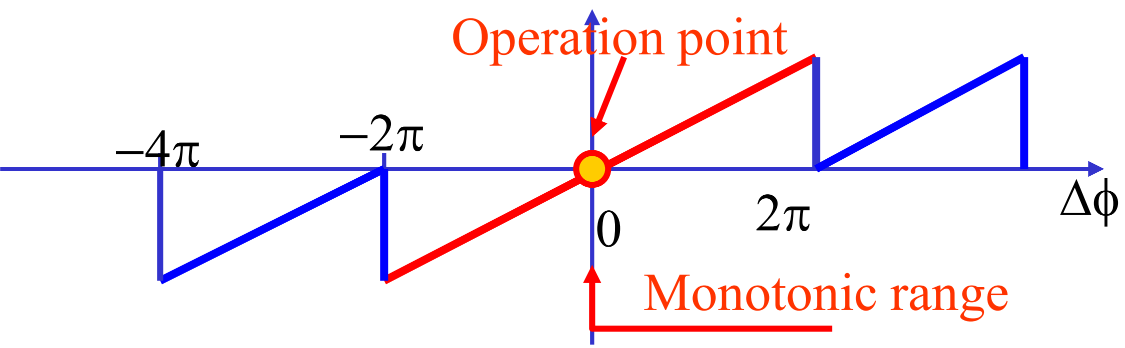


Characteristics of Phase Frequency CPD

- Dead-zone problem

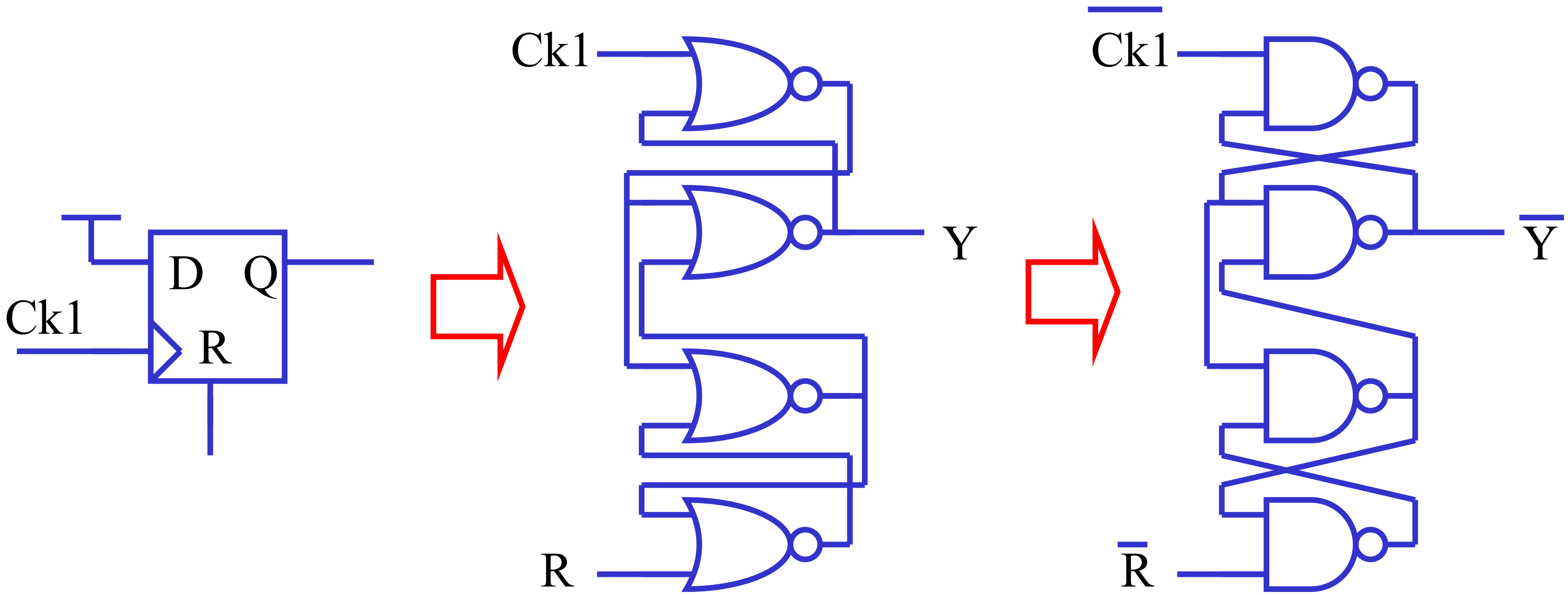






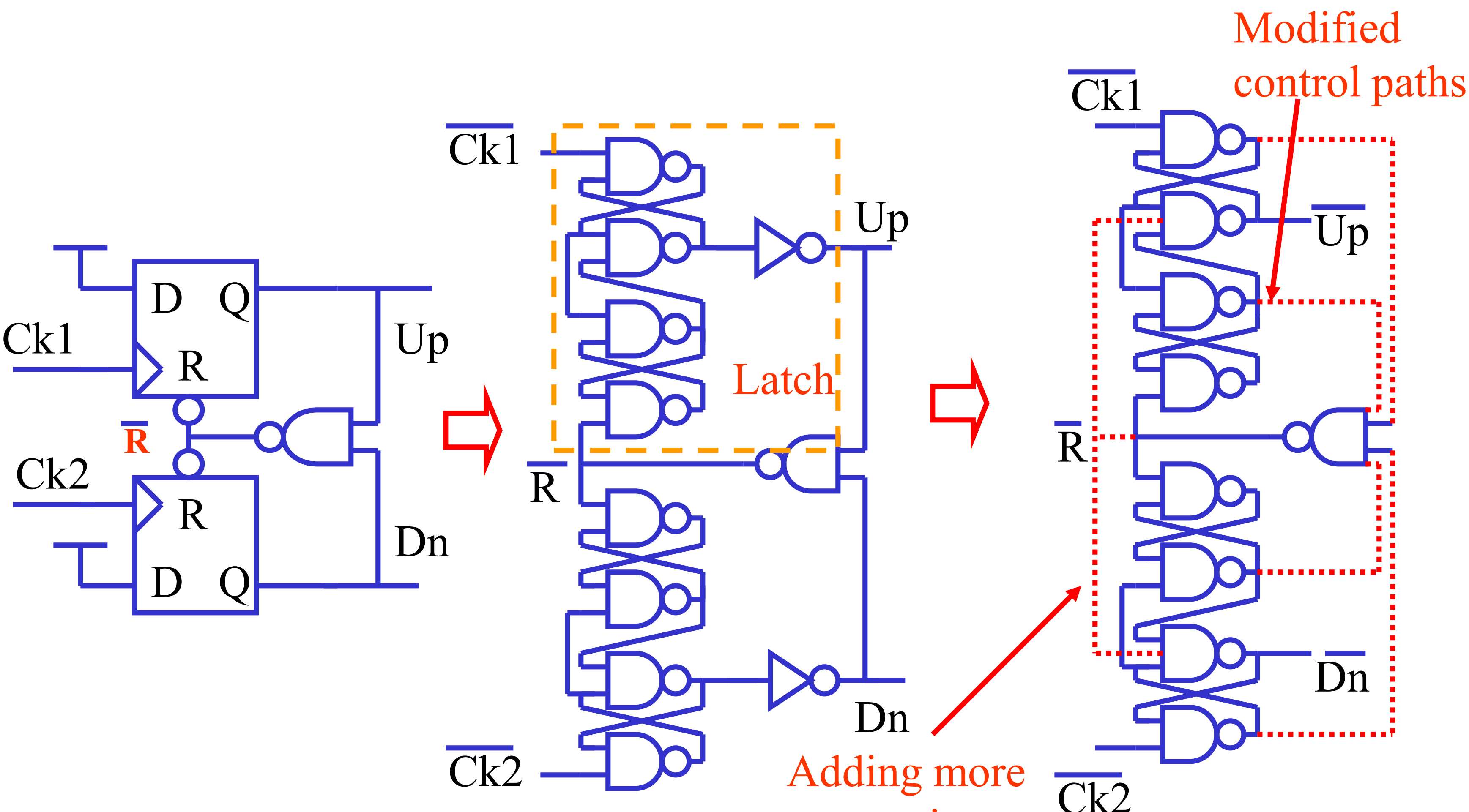
VLSI PFD Circuit Implementation

- VLSI latch circuit



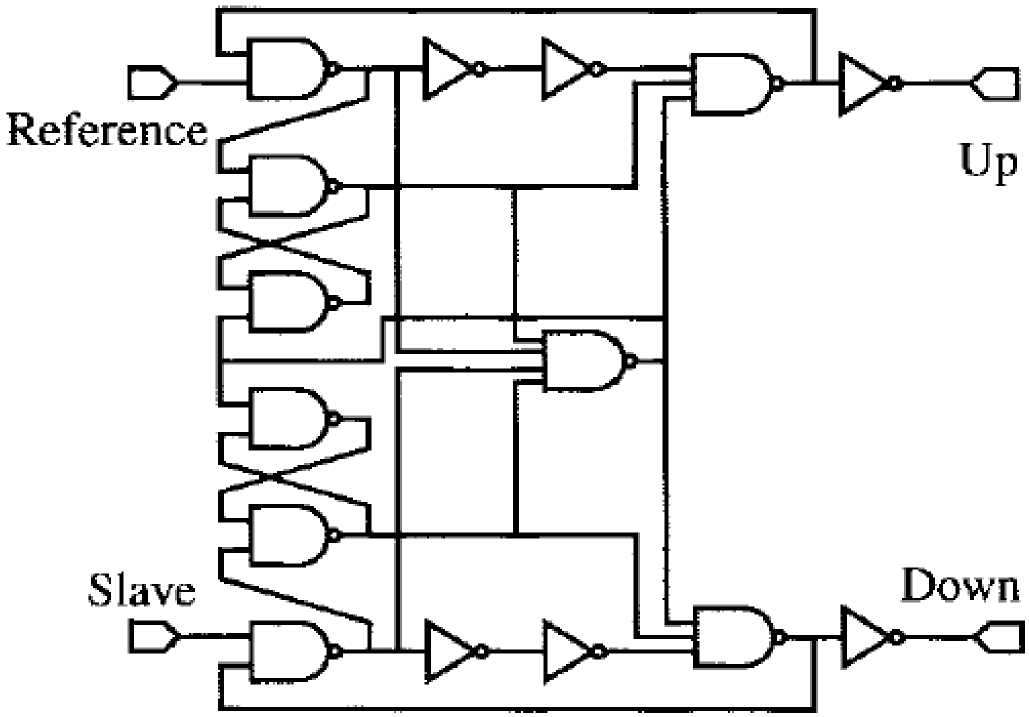
(a) Symbol (b) Nor-based (c) Nand-based

VLSI PFD Circuit Implementation

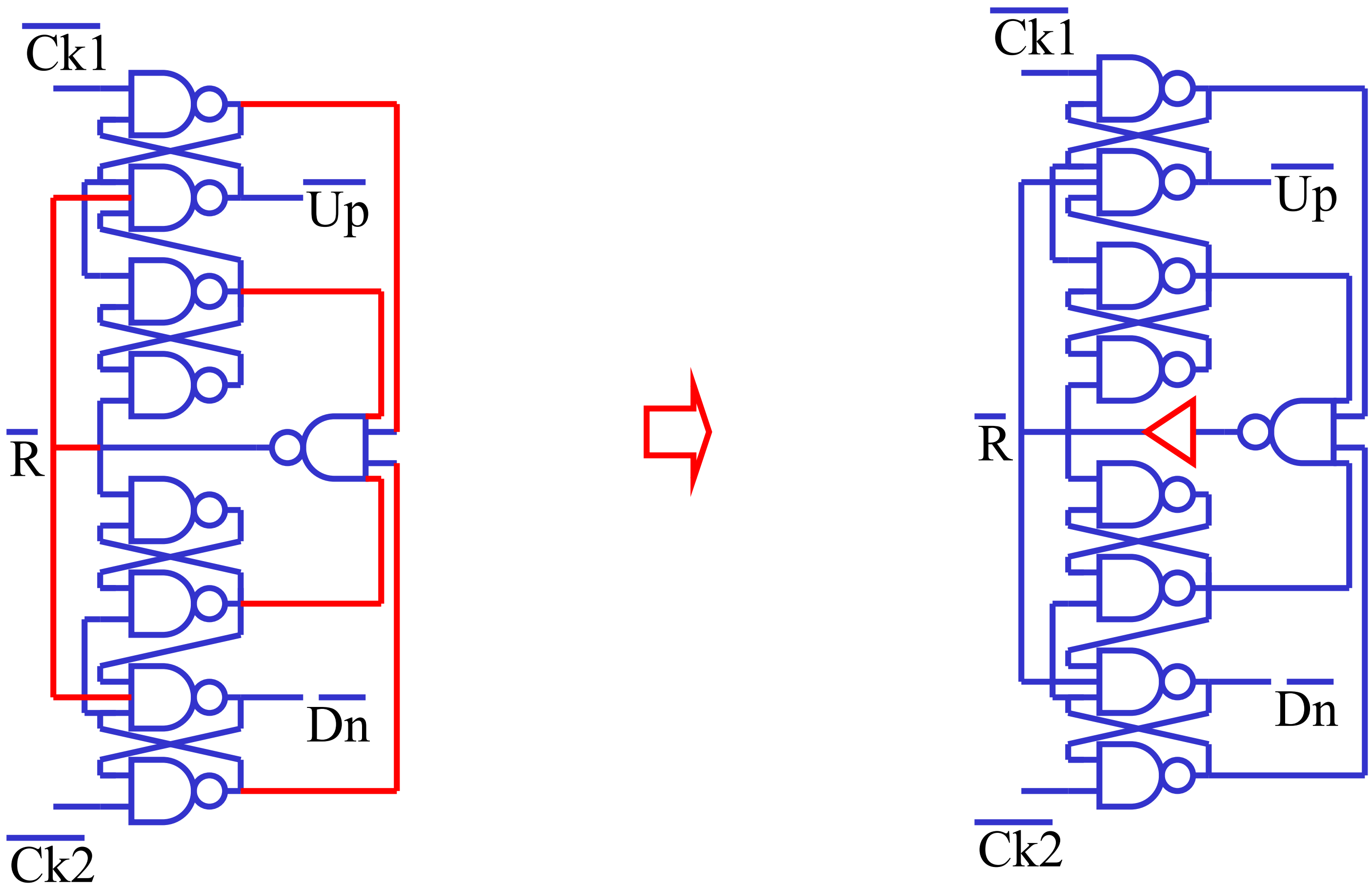


(a) Symbol (b) Basic structure (c) improved

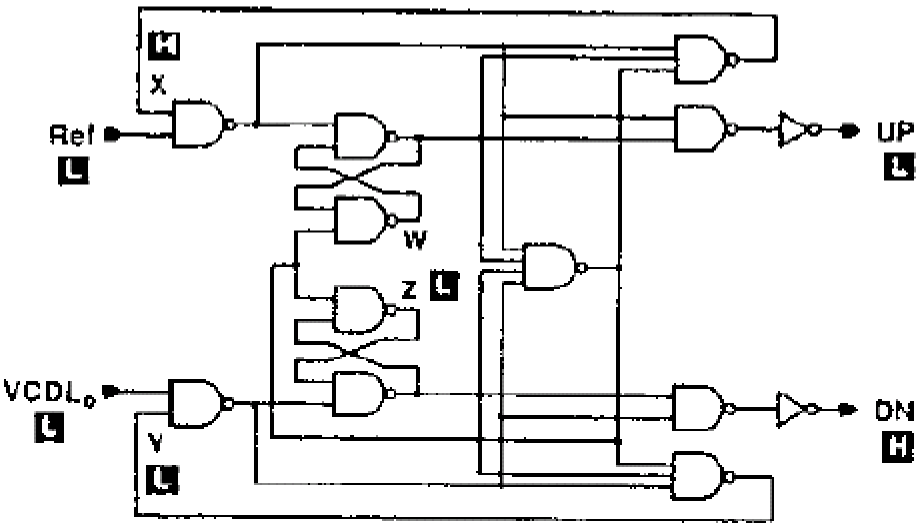
VLSI PFD Circuit Implementations

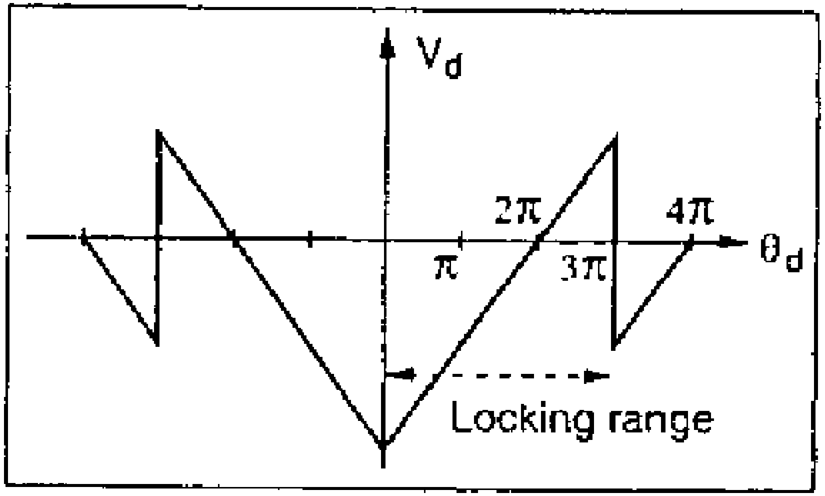


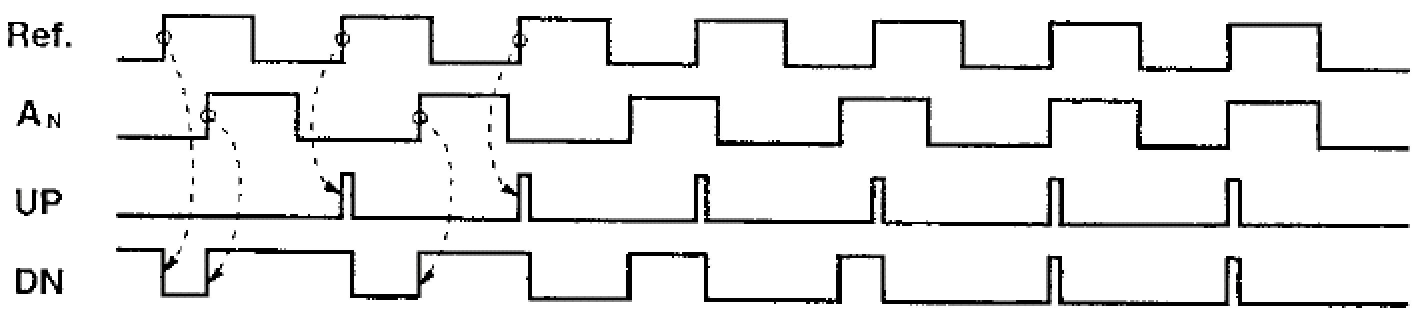
Dead Zone Compensation of PFD



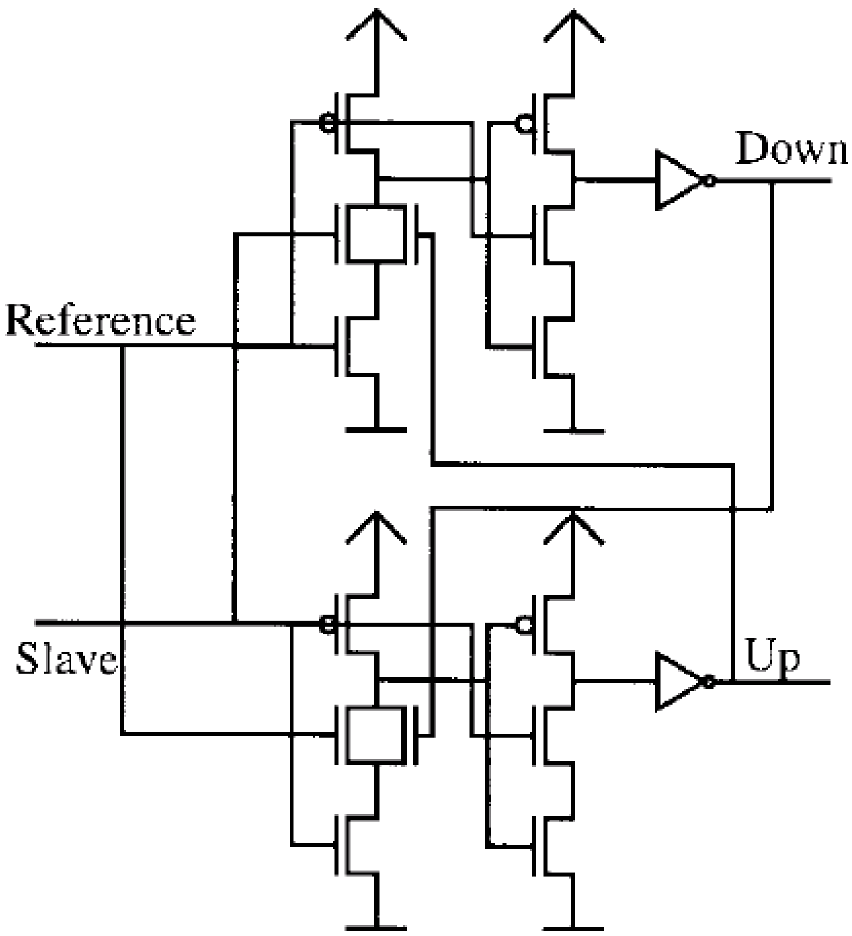
Phase Frequency Detector Circuit







Pre-Charge Type VLSI PFD Circuit



Precharge type phase frequency detector (ptPFD) from [3].

VLSI PD/PFD Circuit Example

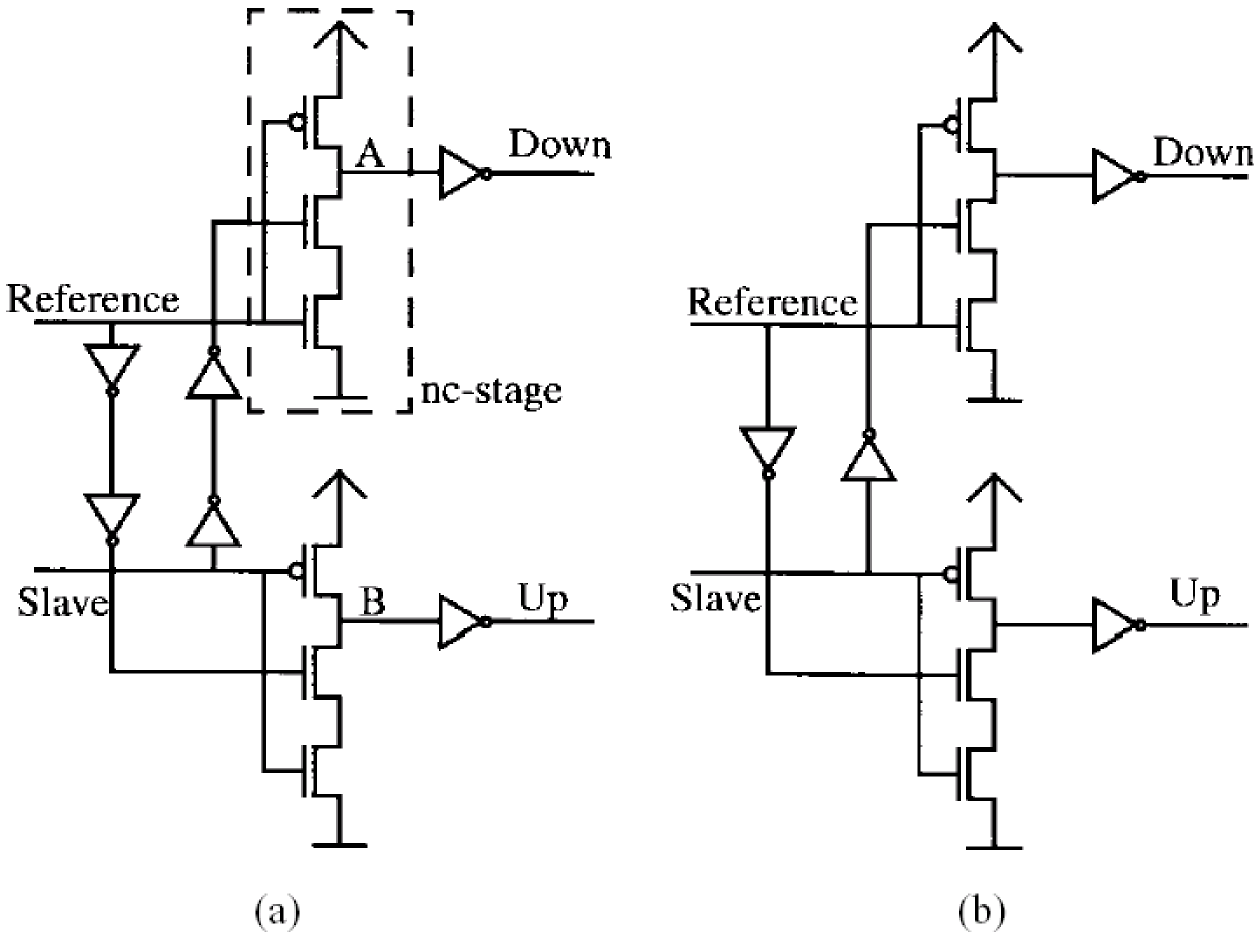
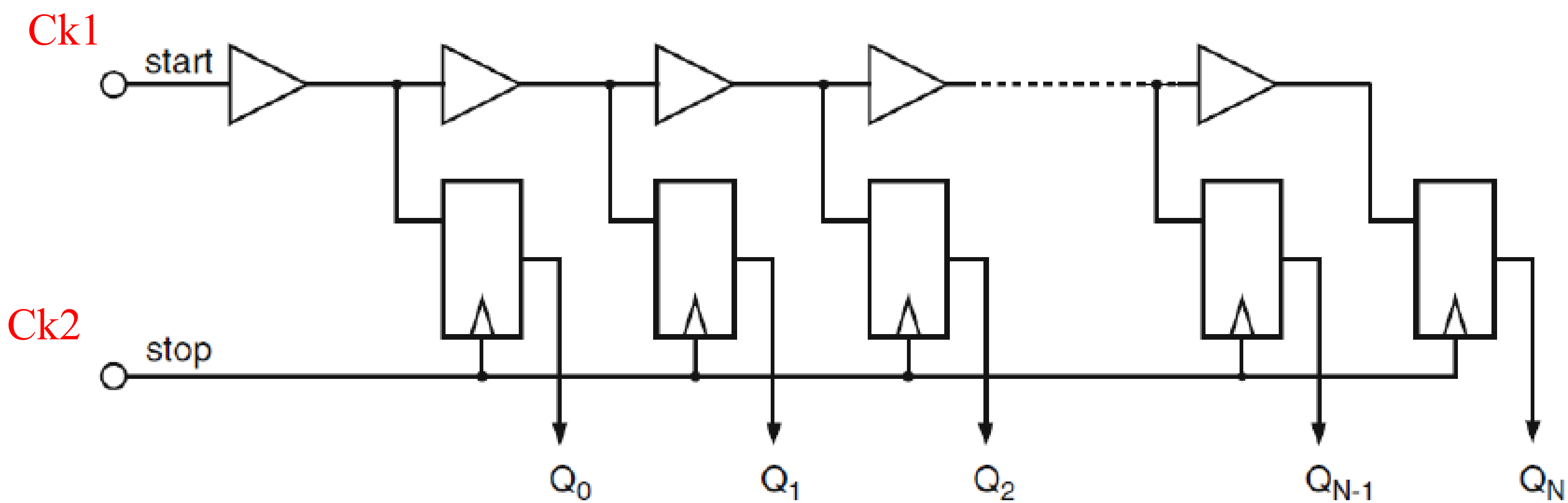
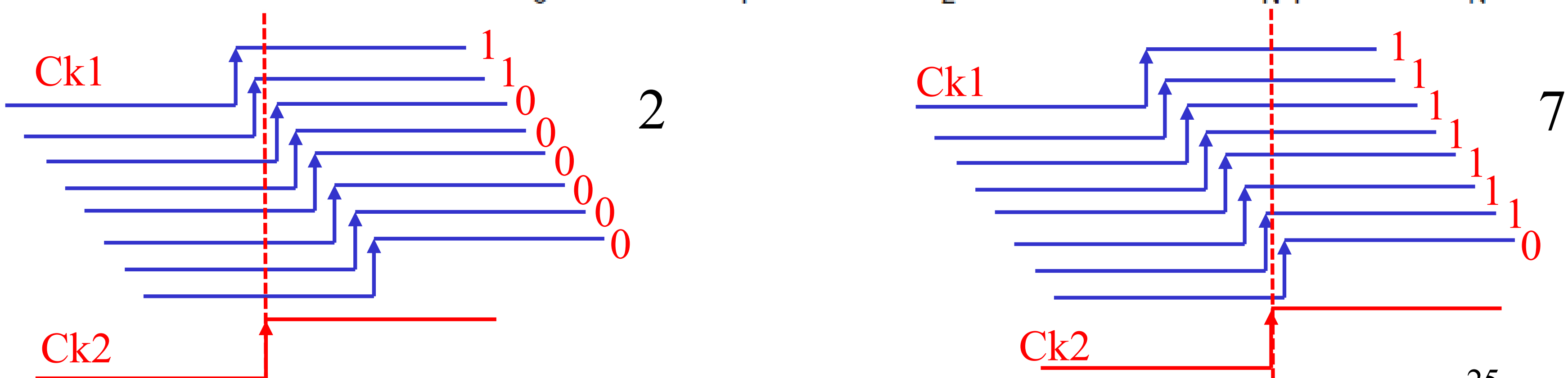


Fig. 3. (a) The ncPFD in zero degree phase offset version. (b) Modified version with π rad phase offset.

VLSI Time-to-Digital Converter (TDC)

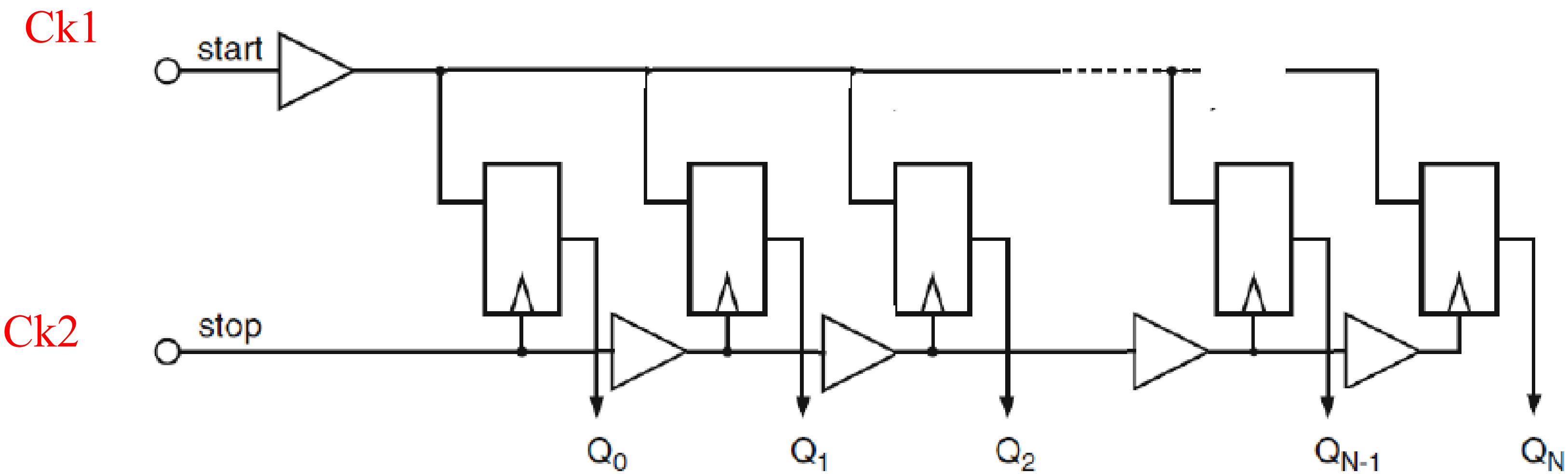
- Basic TDC

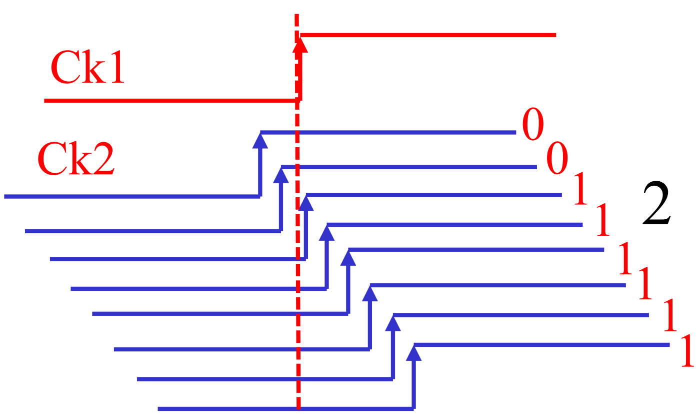


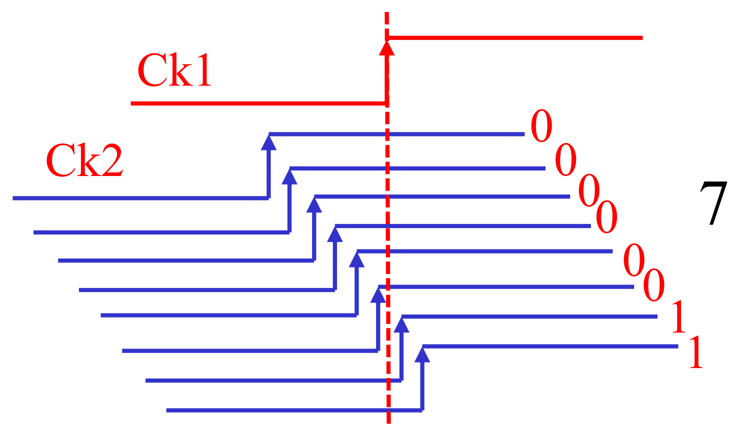


VLSI Time-to-Digital Converter (TDC)

- Oversample TDC

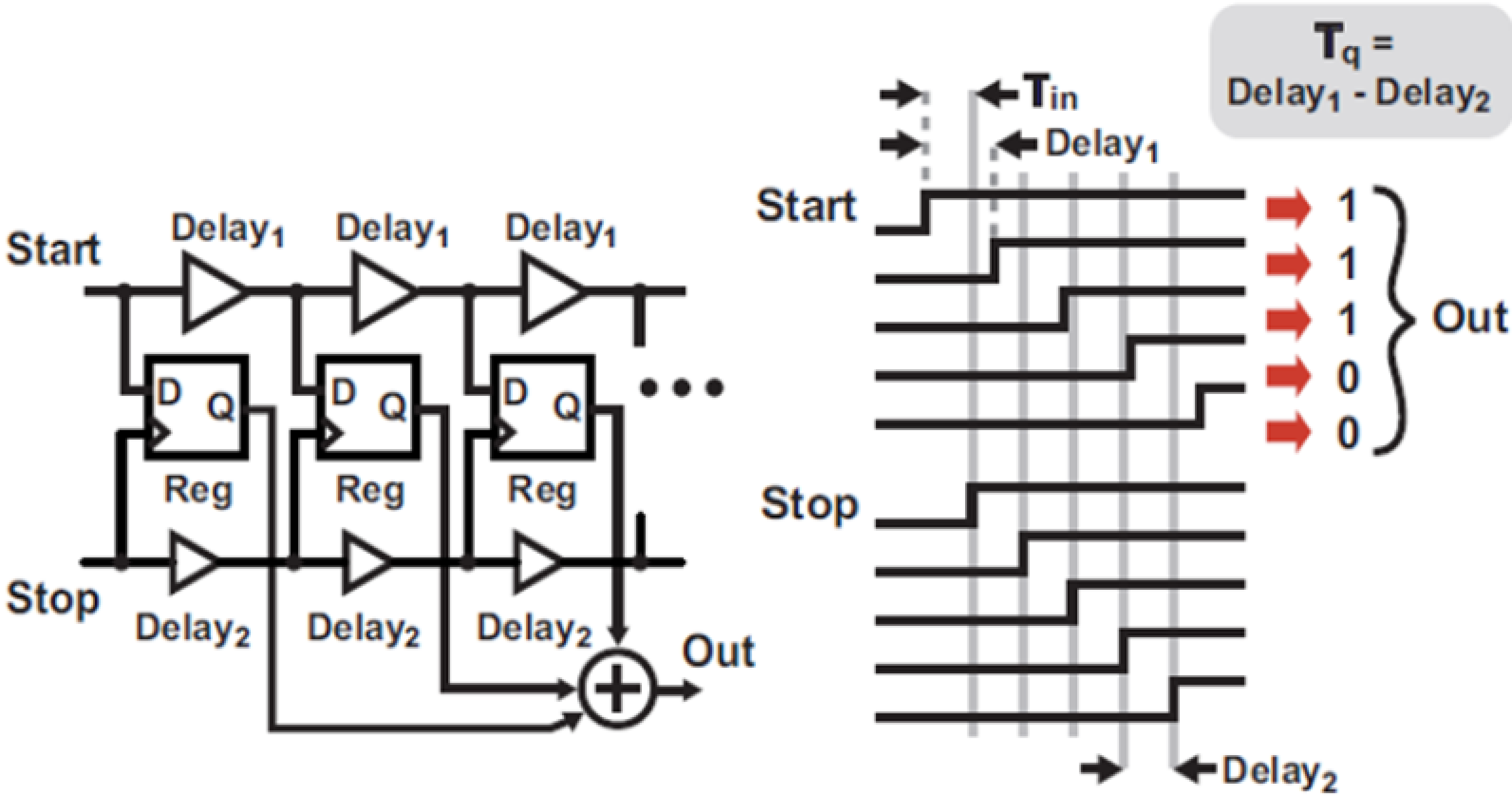






VLSI Time-to-Digital Converter (TDC)

- Improved accuracy TDC



Reading Assignment

- Please read Chapter 9 of the textbook

VLSI High-Speed I/O Circuits

Theoretical Basis, Architecture, Modeling and Circuit Implementation

VLSI High-Speed I/O Circuits

By: Hongjiang Song

ISBN10: 1-4415-5987-6 (Trade Paperback 6x9)
ISBN13: 978-1-4415-5987-6 (Trade Paperback 6x9)
ISBN10: 1-4415-5988-4 (Trade Hardback 6x9)
ISBN13: 978-1-4415-5988-3 (Trade Hardback 6x9)

Pages :	489
Book Format :	Trade Book 6x9
Subject :	TECHNOLOGY / Engineering / Electrical
	TECHNOLOGY / Engineering / General

http://www2.xlibris.com/bookstore/bookdisplay.aspx?bookid=62861